

Instruction for FemSeg_v3_on_50000_sentences_1.ipynb

This document describes step-by-step instructions for running the notebook `FemSeg\_v3\_on\_50000\_sentences\_1.ipynb`, which trains the FEMSeg_kaz_v3 neural segmentation model on a 50,000-sentence corpus and applies it to segment Kazakh text.

1. Purpose of the Notebook

The notebook performs the following tasks:

- training the FEMSeg_kaz_v3 neural model on a CSE-segmented Kazakh corpus (50k sentences);
- saving trained model checkpoints and character vocabularies;
- segmenting a raw Kazakh text file using the trained model;
- converting the segmentation output into SentencePiece-compatible format.

2. Execution Environment

The notebook was developed and tested in Google Colab using Python 3. All input and output files are stored in Google Drive.

3. Mount Google Drive and Set Working Directory

Run the first two cells of the notebook to mount Google Drive and switch to the project directory:

```
from google.colab import drive
drive.mount('/content/drive')

%cd /content/drive/MyDrive/KAZ_MORPH
```

4. Required Input Files

The following files must be present in the directory `/content/drive/MyDrive/KAZ\_MORPH`:

- kazakh_segmented_corpus_50k.txt — CSE-segmented training corpus (50,000 sentences)
- chunk_099_10.txt — raw Kazakh text file for segmentation

If the 50k corpus file does not exist, it can be created from the full corpus using the commented command provided in the notebook.

5. Installing Dependencies

Install the CRF dependency by executing the following cell:

```
!pip install -q pytorch-crf
```

6. Training FEMSeg_kaz_v3

Run the main training cell that defines the FEMSeg architecture, loads the training corpus, and starts model training.

During training, the following files and directories are created:

- `char2id_femseg_v3_50k.json` — character-to-index mapping
- `models_femseg_v3_50k/` — directory containing model checkpoints

7. Segmenting a Text File

After training, the notebook applies the trained model to segment the file ``chunk_099_10.txt``.

The segmentation result is saved as:

``chunk_099_10_femseg_v3_50k.txt``

8. Conversion to SentencePiece Format

The notebook generates a conversion script (``femseg_v3_to_sp.py``) and uses it to convert the segmentation output into SentencePiece-compatible format.

The final output file is:

``chunk_099_10_femseg_v3_sp50.txt``

9. Expected Output Files

After successful execution, the following artifacts should exist in ``/content/drive/MyDrive/KAZ_MORPH``:

- `char2id_femseg_v3_50k.json`
- `models_femseg_v3_50k/`
- `chunk_099_10_femseg_v3_50k.txt`
- `chunk_099_10_femseg_v3_sp50.txt`

10. Reproducibility Notes

The training process uses fixed data and deterministic preprocessing. Although neural training may involve minor stochastic variation, the same configuration was used for all experiments reported in the paper.

11. License and Usage

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